

AmazonHelp AI Support Agent — Report

1. Problem Framing

What “good” means for this brand. AmazonHelp’s public replies are dominated by triage-and-redirect patterns (“please DM us your order ID”) — real resolution happens in private DMs invisible to this dataset. So “good” here means: correct intent, correct routing (risk goes to a human), and a reply that’s honest and on-brand, not necessarily a full public resolution, since that’s rarely how this brand actually operates on this channel. A redirect is a correct outcome, not a failure.

What was chosen not to build:

- **Intent-prefiltered retrieval** — skipped for time/rate-limit reasons; plain semantic search used instead (v1 simplification).
- **Confidence-based routing** — dropped; `classify_intent`’s confidence was uncalibrated (0.92–0.97 even on a flagged borderline case).
- **A trained classifier for production** — real system uses LLM-based classification; keyword/TF-IDF is only the baseline.
- **Full-scope reply-quality eval** — intent/routing use all 150 examples; reply-quality scored on 30, judge-validated on 15 (documented, time-boxed reduction).

2. Results vs. Baselines

Same golden set, same harness, all three tiers.

Intent classification (N=150)

Tier	Accuracy	Macro F1
Trivial (majority class)	41.3%	0.117
Simple (keyword rules)	57.3%	0.465
Real system (LLM)	89.3%	0.894

Clean, large improvement at each tier — the LLM classifier earns its complexity.

Routing (escalate class, N=150)

Trivial’s perfect recall is a false strength — it escalates everything (128 false positives). See Section 4. Real system beats simple, but by a narrower margin than intent classification did.

Tier	Precision	Recall
Trivial (always escalate)	0.15	1.00
Simple (keyword rule)	0.57	0.18
Real system	0.61	0.64

Reply quality (LLM-judge, 1–5, N=30 subset)

Tier	Groundedness	Safety	Tone	Actionability	Overall
Trivial (canned reply)	4.53	4.97	4.50	3.67	4.42
Simple (nearest-neighbor)	2.60	3.47	2.77	2.20	2.76
Real system	4.50	4.67	4.80	4.43	4.60

Real system’s edge is concentrated in actionability — trivial actually edges ahead on groundedness and safety (a vague reply can’t be “wrong”), but loses clearly on actionability, a 0.76-point gap. See Section 4 for why.

3. Failure Analysis — Top 5

1. **Routing misses 36% of real escalations.** 8/150 cases wrongly auto-handled; 3 had explicit “escalate” language in the actual historical reply. Rules only check intent + grounding strength — risk buried in earlier turns slips through.
2. **Reply drafting fabricated an unverified claim.** Thread 1899909 claimed “we’ve escalated this to our technical team” — unsupported by grounding or history (similarity was weak, 0.501). Likely cause: the “give a concrete next step” instruction pushes invention when grounding is thin. Fixed via an explicit anti-fabrication prompt rule.
3. **Nearest-neighbor baseline retrieved unrelated replies.** Threads 226344 and 1355369 scored 1/1/1/1 and 1/1/2/2 — pasted a reply from a genuinely unrelated conversation. Confirms LLM-based adaptation, not just similarity, is doing real work.
4. **“Other” intent leaks into delivery_issue (recall 0.74, weakest class).** 8/39 true-other cases misclassified. Fulfillment errors (“wrong item shipped”) get bucketed as shipping problems — a taxonomy gap, not a model failure.
5. **LLM judge is poorly calibrated on correctness/safety.** Only 53% within-1-point agreement with human scoring (vs. 100% on tone, 87% on groundedness) — weakest on exactly the failure type it’s meant to catch.

4. What Is Misleading About My Headline Number?

- **Overall reply score (4.42 trivial vs. 4.60 real) hides the real gap.** A vague reply can’t be “wrong,” so trivial actually edges ahead on groundedness and safety by default. The real system’s actual edge — 0.76 points higher on actionability (4.43 vs. 3.67) — is buried in the average.
- **89.3% intent accuracy hides its weakest class.** “Other,” the catch-all safety net, has the lowest recall (0.74).

- **Reply-quality scores are only as reliable as the judge’s weakest dimension.** Correctness/safety agreement is 53% — the dimension tied directly to fabrication risk.
- **Sample sizes shrink at each stage.** N=150 (intent/routing) → N=30 (reply quality) → N=15 (judge validation) — all documented, time-boxed reductions. Treat later numbers with wider uncertainty.

5. What I’d Do With One More Week

- Fix routing recall with a second-pass check for context-buried risk signals.
- Re-run reply-quality eval on the full 150, not a 30-row subset.
- Add a dedicated correctness/safety verification pass, separate from the general judge.
- Re-introduce intent-prefiltered retrieval and measure its actual effect.
- Split the delivery_issue/other taxonomy boundary to capture fulfillment errors separately.
- Expand judge-validation to 40+ examples for a tighter agreement estimate.
- Test whether the anti-fabrication prompt rule generalizes beyond thread 1899909.

See `decision_log.md` in the repository for the full list of 15 non-obvious decisions made during this project, with rationale for each.